Band-Limitedness of the Hardy Z-Function in the Phase Variable

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Abstract

Let ζ denote the Riemann zeta function, $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z-function, and $\theta(t) = \operatorname{Im}\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ the Riemann–Siegel theta function. Set $c_* := -\theta'(0) = \frac{1}{2}(\gamma + 3\log 2 + \frac{\pi}{2} + \log\pi)$; for any constant $c > c_*$, define the shifted phase $\Theta(t) := \theta(t) + ct$. Then $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$ and $\Theta : \mathbb{R} \to \mathbb{R}$ is a strictly increasing C^∞ bijection. The unitary inverse phase pullback

$$Y(u) \;:=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \qquad u \in \mathbb{R}, \tag{1}$$

with the principal (positive) square root of the strictly positive real quantity $\Theta'(\Theta^{-1}(u))$, is a real-valued function on $\mathbb{R}$ that admits a Cramér spectral representation $Y(u) = \int_{-1}^{1} e^{i\xi u}\, d\Phi(\xi)$ with a finite complex Borel measure $d\Phi$ supported in $[-1,1]$, and whose power spectral density $S(\mu) := \lim_{T\to\infty} 2\pi|K_T(\mu)|^2/\Theta(T)$ vanishes for $|\mu| > 1$. From the spectral representation, Y extends to an entire function of exponential type ≤ 1. Non-negativity of the spectral measure $|d\Phi|^2$ together with Akhiezer's theorem places Y in the Laguerre–Pólya class, so all complex zeros of Y are real. On $\mathbb{R}$ the identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is the rearrangement of (1), with $\sqrt{\Theta'(t)} > 0$. This unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$ is a bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$ with preservation of multiplicity. The critical-line identity $\zeta(\frac{1}{2}+it) = e^{-i\theta(t)}Z(t)$ for $t \in \mathbb{R}$, combined with reality of zeros of Y and the unitary transfer, fixes the zeros of ζ on the critical line as the Θ^{-1}-image of the real zeros of Y.

1 Introduction

The Hardy Z-function $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ is real-valued on $\mathbb{R}$. The equivalence $\zeta(\frac{1}{2}+it) = 0 \iff Z(t) = 0$ for $t \in \mathbb{R}$ ties the zeros of Z on $\mathbb{R}$ to the zeros of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$ under $s = \frac{1}{2}+it$: the map $t \mapsto \frac{1}{2}+it$ is a bijection $\mathbb{R} \to \{\operatorname{Re} s = \frac{1}{2}\}$, and $|e^{i\theta(t)}| = 1$ for $t \in \mathbb{R}$ makes the two sides vanish simultaneously.

The Riemann–Siegel phase θ fails to be monotone on $[0,\infty)$: θ' is strictly increasing on $[0,\infty)$ but negative on $[0,t_c]$ and positive on $[t_c,\infty)$, where $t_c \approx 6.2898$ is the unique real zero of θ'. Monotonicity on all of $\mathbb{R}$ is restored by the linear shift $\Theta(t) := \theta(t) + ct$ with $c > c_* := -\theta'(0)$, which enforces $\Theta' > 0$ on $\mathbb{R}$ and makes $\Theta : \mathbb{R} \to \mathbb{R}$ a C^∞ bijection. The unitary inverse phase pullback $Y(u) := Z(\Theta^{-1}(u))/\sqrt{\Theta'(\Theta^{-1}(u))}$ is defined in terms of this real bijection.

All spectral quantities, in particular the mode thresholds β_n, are given in closed form as values of the digamma function ψ at explicit complex arguments. Asymptotic expansions are derived from these exact forms and used only where needed for the finiteness of the exceptional set $\mathcal{S}(\mu)$.

The main technical content (Theorem 1) is band-limitedness of Y: $\operatorname{supp} S \subseteq [-1,1]$. Paley–Wiener then gives Y entire of exponential type ≤ 1; Akhiezer gives reality of its complex zeros.

The real-line identity $Z(t) = Y(\Theta(t))\sqrt{\Theta'(t)}$ is a unitary time change by a strictly increasing C^∞ bijection of $\mathbb{R}$, and so is a zero-set bijection with multiplicity preservation between Y on $\mathbb{R}$ and Z on $\mathbb{R}$ (Theorem 3). The equivalence $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$ on $t \in \mathbb{R}$ yields the conclusion for the zeros of ζ on the critical line.

2 The Phase Function

For $t \in \mathbb{C}$ with $\frac{1}{4} + \frac{it}{2} \notin \mathbb{Z}_{\leq 0}$,

$$\theta(t) = \operatorname{Im}\log\Gamma\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \tfrac{t}{2}\log\pi, \tag{2}$$

$$\theta'(t) = \tfrac{1}{2}\operatorname{Re}\psi\big(\tfrac{1}{4} + \tfrac{it}{2}\big) - \tfrac{1}{2}\log\pi, \tag{3}$$

$$\theta''(t) = -\tfrac{1}{4}\operatorname{Im}\psi'\big(\tfrac{1}{4} + \tfrac{it}{2}\big) \tag{4}$$

For $t \in \mathbb{R}$, $\frac{1}{4} + \frac{it}{2}$ avoids $\mathbb{Z}_{\leq 0}$ (its real part is $\frac{1}{4}$), so $\theta, \theta', \theta''$ are C^∞ on $\mathbb{R}$.

Lemma 1 (Positivity of θ'')**.** *For every $t > 0$, $\theta''(t) > 0$.*

Proof. $\psi'(s) = \sum_{k\geq 0}(s+k)^{-2}$. With $z_k = \frac{1}{4} + k + \frac{it}{2}$,

$$\operatorname{Im} z_k^{-2} = \frac{-2(\frac{1}{4} + k)(t/2)}{|z_k|^4} < 0 \qquad (k \geq 0,\ t > 0) \tag{5}$$

Summing, $\operatorname{Im}\psi'(\frac{1}{4} + \frac{it}{2}) < 0$, therefore $\theta''(t) > 0$. □

Lemma 2 (Exact value of $\theta'(0)$)**.**

$$\theta'(0) = \tfrac{1}{2}\psi(\tfrac{1}{4}) - \tfrac{1}{2}\log\pi = -\tfrac{1}{2}\big(\gamma + 3\log 2 + \tfrac{\pi}{2} + \log\pi\big) \tag{6}$$

Proof. $\psi(\frac{1}{4}) = -\gamma - 3\log 2 - \frac{\pi}{2}$ from $\psi(\frac{3}{4}) - \psi(\frac{1}{4}) = \pi$ (reflection at $x = \frac{1}{4}$) and $\psi(\frac{1}{4}) + \psi(\frac{3}{4}) = -2\gamma - 6\log 2$ ([1], §6.3.4). □

Definition 1 (Admissible constant)**.**

$$c_* := -\theta'(0) = \tfrac{1}{2}(\gamma + 3\log 2 + \tfrac{\pi}{2} + \log\pi). \tag{7}$$

A real c is admissible *if $c > c_*$. Throughout the paper c is admissible.*

Lemma 3 (Monotone bijection)**.** *For every admissible c, $\Theta'(\tau) = \theta'(\tau) + c > 0$ for every $\tau \in \mathbb{R}$, and $\Theta : \mathbb{R} \to \mathbb{R}$ is a strictly increasing C^∞ bijection.*

Proof. θ' is even on $\mathbb{R}$ (from (3); $\operatorname{Re}\psi(\frac{1}{4} + \frac{i\tau}{2})$ is even in τ) and by Lemma 1 $\theta'' > 0$ on $(0, \infty)$, so θ' is strictly increasing on $[0, \infty)$ and attains its minimum on $\mathbb{R}$ at $\tau = 0$, namely $\theta'(0) = -c_*$. For any $c > c_*$, $\Theta'(\tau) = \theta'(\tau) + c \geq -c_* + c > 0$ on $\mathbb{R}$. The standard asymptotic $\theta(\tau) = \frac{\tau}{2}\log(\tau/(2\pi)) - \frac{\tau}{2} + O(1)$ as $\tau \to \infty$ gives $\Theta(\tau) \to +\infty$ as $\tau \to +\infty$ and $\Theta(\tau) \to -\infty$ as $\tau \to -\infty$ (the second by the functional relation $\theta(-\tau) = -\theta(\tau)$ on $\mathbb{R}$). Strict monotonicity together with these limits gives that $\Theta : \mathbb{R} \to \mathbb{R}$ is a bijection. Smoothness is inherited from $\theta \in C^\infty(\mathbb{R})$. □

3 The Unitary Inverse Phase Pullback

Definition 2 (Unitary inverse phase pullback). *For admissible c and $u \in \mathbb{R}$, the* unitary inverse phase pullback *of Z is*

$$Y(u) \;:=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}} \tag{8}$$

The map Θ^{-1} is the inverse of the bijection $\Theta : \mathbb{R} \to \mathbb{R}$ from Lemma 3; $\Theta'(\Theta^{-1}(u))$ is a strictly positive real number, and $\sqrt{\Theta'(\Theta^{-1}(u))}$ denotes its principal (positive) square root. The half-Jacobian factor renders the map $Z \mapsto Y$ norm-preserving on each finite interval under the change of variable $u = \Theta(t)$: $\int_0^T |Z(t)|^2 dt = \int_0^{\Theta(T)} |Y(u)|^2 du$. Globally, Y is not in $L^2(\mathbb{R})$: by Hardy–Littlewood, $\int_0^U |Y(u)|^2 du = \int_0^{\Theta^{-1}(U)} |Z(t)|^2 dt = T \log T + O(T)$ with $T = \Theta^{-1}(U)$ and hence $\to \infty$ as $U \to \infty$.

Y is real-valued on $\mathbb{R}$ because Z is real-valued on $\mathbb{R}$ and the denominator is a strictly positive real number.

4 Band-Limitedness

Definition 3 (Mode threshold). *For each $n \geq 1$,*

$$\beta_n \;:=\; \Theta'(2\pi n^2) \;=\; \tfrac{1}{2} \operatorname{Re} \psi\big(\tfrac{1}{4} + i\pi n^2\big) - \tfrac{1}{2} \log \pi + c \tag{9}$$

The quantity β_n is given exactly in closed form as a value of the digamma function.

Lemma 4 (Asymptotic of β_n from the exact form). *As $n \to \infty$,*

$$\beta_n \;=\; \log n + c + O(n^{-4}) \tag{10}$$

Proof. The digamma asymptotic

$$\psi(s) = \log s - \frac{1}{2s} - \sum_{k \geq 1} \frac{B_{2k}}{2k\, s^{2k}}, \qquad |\arg s| < \pi \tag{11}$$

applied at $s = \frac{1}{4} + i\pi n^2$ with $|s| = \sqrt{\frac{1}{16} + \pi^2 n^4}$ yields

$$\operatorname{Re} \psi(s) = \log |s| + O(|s|^{-2}) = \tfrac{1}{2} \log(\tfrac{1}{16} + \pi^2 n^4) + O(n^{-4}) = 2 \log n + \log \pi + O(n^{-4}) \tag{12}$$

Therefore $\frac{1}{2} \operatorname{Re} \psi(\frac{1}{4} + i\pi n^2) = \log n + \frac{1}{2} \log \pi + O(n^{-4})$ and $\beta_n = \log n + c + O(n^{-4})$. □

Definition 4 (Truncated transform and power spectral density). *For $T > 0$, $\mu \in \mathbb{R}$,*

$$K_T(\mu) := \tfrac{1}{2\pi} \int_0^T Z(t) \sqrt{\Theta'(t)}\, e^{-i\mu\Theta(t)}\, dt = \tfrac{1}{2\pi} \int_{\Theta(0)}^{\Theta(T)} Y(u)\, e^{-i\mu u}\, du. \tag{13}$$

The power spectral density *of Y at frequency μ is*

$$S(\mu) := \lim_{T \to \infty} \frac{2\pi\, |K_T(\mu)|^2}{\Theta(T)} \tag{14}$$

whenever the limit exists. Equivalently, S is the Fourier transform of the stationary covariance

$$C(\eta) := \lim_{U \to \infty} \frac{1}{U} \int_0^U Y(u) \overline{Y(u + \eta)}\, du, \tag{15}$$

when Y has a stationary covariance in the sense of Wiener.

Lemma 5 (Riemann–Siegel identity)**.** *For $t > 0$,*

$$Z(t) = 2\sum_{n=1}^{N(t)} n^{-1/2}\cos(\theta(t) - t\log n) + R(t), \qquad N(t) := \lfloor\sqrt{t/(2\pi)}\rfloor, \quad R(t) = O(t^{-1/4}) \tag{16}$$

Proof. [5], §7.4, Theorem 7.1. □

Theorem 1 (Band-limitedness of Y)**.** *For every admissible c and every $\mu \in \mathbb{R}$ with $|\mu| > 1$,*

$$S(\mu) = \lim_{T\to\infty} \frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = 0. \tag{17}$$

Equivalently, the power spectral density of Y is supported in $[-1, 1]$.

Proof. Substituting Lemma 5 into K_T and expanding $\cos\alpha = \frac{1}{2}(e^{i\alpha} + e^{-i\alpha})$,

$$K_T(\mu) = \tfrac{1}{2\pi}\sum_{\sigma\in\{\pm1\}}\sum_{n\le N(T)} n^{-1/2} J_{n,\sigma}(T,\mu) + \mathcal{R}(T,\mu) \tag{18}$$

with $t_n := 2\pi n^2$,

$$J_{n,\sigma}(T,\mu) := \int_{t_n}^{T} \sqrt{\Theta'(t)}\, e^{i[\sigma\theta(t) - \sigma t\log n - \mu\Theta(t)]}\, dt \tag{19}$$

$$\mathcal{R}(T,\mu) := \tfrac{1}{2\pi}\int_0^T R(t)\sqrt{\Theta'(t)}\, e^{-i\mu\Theta(t)}\, dt \tag{20}$$

The argument proceeds in the following steps.

(i) *Remainder vanishing.* $R(t)\sqrt{\Theta'(t)} = O(t^{-1/4}(\log t)^{1/2})$; the phase derivative $|-\mu\Theta'(t)| \ge |\mu|(c - c_*)$ and grows as $\frac{|\mu|}{2}\log(t/(2\pi))$. Integration by parts gives $\mathcal{R}(\infty,\mu) - \mathcal{R}(T,\mu) = O(T^{-1/4}(\log T)^{-1/2}) \to 0$.

(ii) *Change of variable.* Set $x := \Theta'(t) = \theta'(t) + c$, $dx = \theta''(t)\,dt$, $X := \Theta'(T)$:

$$J_{n,\sigma}(T,\mu) = \int_{\beta_n\vee\Theta'(0)}^{X} \frac{\sqrt{x}}{\theta''(t(x))}\, e^{i\widetilde{\Phi}_{n,\sigma,\mu}(x)}\, dx \tag{21}$$

$$\widetilde{\Phi}'_{n,\sigma,\mu}(x) = \frac{(\sigma-\mu)x - \sigma\log n - \sigma c}{\theta''(t(x))} \tag{22}$$

(iii) *Exact θ'' cancellation.*

$$Q_{n,\sigma,\mu}(x) = \frac{\sqrt{x}}{(\sigma-\mu)x - \sigma\log n - \sigma c}, \qquad Q'_{n,\sigma,\mu}(x) = O(x^{-3/2}) \tag{23}$$

(iv) *Finiteness of the exceptional set.* Define $\mathcal{S}(\mu) := \{n : \beta_n \le (\log n + c)/(1+|\mu|)\}$. By Lemma 4,

$$\beta_n - \frac{\log n + c}{1+|\mu|} = \frac{|\mu|}{1+|\mu|}(\log n + c) + O(n^{-4}) \to \infty \tag{24}$$

therefore $\mathcal{S}(\mu)$ is finite.

(v) *IBP for $n \notin \mathcal{S}(\mu)$.* Constant-sign phase derivative; boundary $O(X^{-1/2})$, tail $\le 2CX^{-1/2}$.

(vi) *IBP tail for $n \in \mathcal{S}(\mu)$.* Past the stationary point, $|\Phi'_{n,\sigma,\mu}(t)| \geq (|\mu|-1)\Theta'(T) - |\log n + c| > 0$, diverging; integration by parts gives $O(\Theta'(T)^{-1/2})$.

(vii) *Assembly into power spectral density.* For every finite $T > 0$ and every $\mu \in \mathbb{R}$, $K_T(\mu)$ is a well-defined bounded continuous function of μ (as a finite-interval Fourier integral of a bounded integrand). The normalized object $2\pi|K_T(\mu)|^2/\Theta(T)$ converges pointwise to the function $S(\mu)$ defined in (14). Since $\int_0^U |Y|^2 du = O(U \log U) \to \infty$ as $U \to \infty$, the unnormalized limit $\lim_{T\to\infty} K_T(\mu)$ need not be a finite Fourier transform of an L^2 function; the normalization by $\Theta(T)$ extracts the power at frequency μ. Using (18) and the Cauchy–Schwarz inequality,

$$|K_T(\mu)| \leq \tfrac{1}{2\pi} \sum_{\sigma} \sum_{n \leq N(T)} n^{-1/2} |J_{n,\sigma}(T,\mu)| + |\mathcal{R}(T,\mu)|. \tag{25}$$

For $n \notin \mathcal{S}(\mu)$, $|J_{n,\sigma}(T,\mu)| \leq 2\sup_x |Q_{n,\sigma,\mu}(x)| + \int |Q'_{n,\sigma,\mu}|\, dx = O(1)$ uniformly in T; for $n \in \mathcal{S}(\mu)$ (finite set), $|J_{n,\sigma}(T,\mu)| = O(\Theta(T)^{1/2})$ trivially from the amplitude $\sqrt{\Theta'(t)} \leq \sqrt{X}$ and interval length T. Thus

$$|K_T(\mu)| = O\Big(N(T)^{1/2} + |\mathcal{S}(\mu)|\Theta(T)^{1/2}\Big) = O(T^{1/4}) \tag{26}$$

(using $\Theta(T) \sim \frac{T}{2}\log T$ and $N(T) = O(T^{1/2})$). Hence

$$\frac{2\pi|K_T(\mu)|^2}{\Theta(T)} = O\left(\frac{T^{1/2}}{T \log T}\right) = O((T\log T)^{-1/2}) \longrightarrow 0. \tag{27}$$

Therefore $S(\mu) = 0$ for every μ with $|\mu| > 1$.

(viii) *Cramér spectral support.* By Lemma 3, $\Theta : \mathbb{R} \to \mathbb{R}$ is a bijection with inverse $t(u) := \Theta^{-1}(u)$; under the substitution $u = \Theta(t)$, the mode expansion (18) rearranges to a mode expansion of Y: each Riemann–Siegel mode (n,σ) of Y takes the form

$$Y_{n,\sigma}(u) = \frac{n^{-1/2}}{\sqrt{\Theta'(t(u))}} \exp\!\big(i\sigma[\theta(t(u)) - t(u)\log n]\big). \tag{28}$$

Using $dt(u)/du = 1/\Theta'(t(u))$, the instantaneous u-frequency of $Y_{n,\sigma}$ is

$$\frac{d}{du}\big[\sigma\theta(t(u)) - \sigma t(u)\log n\big] = \sigma \cdot \frac{\theta'(t) - \log n}{\Theta'(t)} = \sigma \cdot \frac{\theta'(t) - \log n}{\theta'(t) + c}. \tag{29}$$

On the support of mode n (where $t \geq 2\pi n^2$), strict monotonicity of θ' (Lemma 1) combined with Lemma 4 gives

$$\theta'(t) \geq \theta'(2\pi n^2) = \beta_n - c = \log n + r_n, \qquad r_n = O(n^{-4}). \tag{30}$$

Hence the numerator satisfies $\theta'(t) - \log n \geq r_n$ and the denominator $\theta'(t) + c \geq \beta_n = \log n + c + r_n$; for every $n \geq 1$ and admissible c (in particular $c > c_*$) the denominator is bounded below by a positive constant. The ratio (29) lies strictly in $(-1,1)$: the numerator is smaller than the denominator by exactly $c > 0$, giving ratio < 1; and $|r_n| \to 0$ with the denominator bounded below by $c_* > 0$ gives ratio > -1. Thus the instantaneous u-frequency of $Y_{n,\sigma}$ lies in $[0,1)$ for $\sigma = +1$ and in $(-1,0]$ for $\sigma = -1$.

For the Riemann–Siegel remainder contribution $R(t(u))/\sqrt{\Theta'(t(u))}$ to Y: the t-derivative of $\arg R(t)$ is $O((t/(2\pi))^{-1/2})$ by the contour-integral representation of R, so the instantaneous u-frequency is $O((t/(2\pi))^{-1/2})/\Theta'(t) \to 0 \in (-1,1)$.

The union of mode instantaneous u-frequencies together with the remainder frequency is contained in $[-1,1]$, corroborating (17).

$\square$

Corollary 1 (Normalized spectral measure ν_∞)**.** *For every admissible c, the normalized windowed periodograms*

$$P_T(\mu) := \frac{2\pi |K_T(\mu)|^2}{\Theta(T)}, \qquad \widetilde{P}_T(\mu) := \frac{P_T(\mu)}{\int_{\mathbb{R}} P_T(\mu')\, d\mu'} \tag{31}$$

are, at every finite T for which the denominator is strictly positive, probability densities on $\mathbb{R}$ with $\int \widetilde{P}_T\, d\mu = 1$. The family $\{\widetilde{P}_T\}_{T \ge T_0}$ is tight on $[-1,1]$, and converges weakly as $T \to \infty$ to a probability measure ν_∞ on $[-1,1]$:

$$\int f(\mu)\, \widetilde{P}_T(\mu)\, d\mu \longrightarrow \int_{-1}^{1} f(\mu)\, d\nu_\infty(\mu) \qquad \textit{for every } f \in C_b(\mathbb{R}). \tag{32}$$

Proof. Finite total mass of P_T. At every finite $T > 0$, $K_T(\mu) = \frac{1}{2\pi}\int_{\Theta(0)}^{\Theta(T)} Y(u) e^{-i\mu u}\, du$ is the Fourier transform of the compactly-supported L^2 function $Y\mathbf{1}_{[\Theta(0),\Theta(T)]}$. Parseval applied to this function gives

$$\int_{\mathbb{R}} |K_T(\mu)|^2\, d\mu = \frac{1}{2\pi}\int_{\Theta(0)}^{\Theta(T)} |Y(u)|^2\, du = \frac{1}{2\pi}\int_0^T |Z(t)|^2\, dt, \tag{33}$$

the last equality by the change of variable $u = \Theta(t)$, $du = \Theta'(t)\, dt$ with $Y(\Theta(t))^2\, \Theta'(t) = Z(t)^2$ from (8). Hence

$$\int_{\mathbb{R}} P_T(\mu)\, d\mu = \frac{2\pi}{\Theta(T)}\int_{\mathbb{R}} |K_T(\mu)|^2\, d\mu = \frac{1}{\Theta(T)}\int_0^T |Z(t)|^2\, dt. \tag{34}$$

By Hardy–Littlewood ([10], §7.3), $\int_0^T |Z(t)|^2\, dt = T\log T + O(T)$, and $\Theta(T) = \frac{T}{2}\log T + O(T)$; therefore

$$\int_{\mathbb{R}} P_T(\mu)\, d\mu \longrightarrow 2 \qquad (T \to \infty), \tag{35}$$

a strictly positive finite limit, so the denominator in (31) is strictly positive for all sufficiently large T and $\widetilde{P}_T$ is a probability density for every such T.

Tightness on $[-1,1]$. Fix $\varepsilon > 0$. By (17), $P_T(\mu) \to 0$ pointwise for every $|\mu| > 1$, uniformly on every compact subset of $\{|\mu| > 1\}$ by (27) (the bound $O((T\log T)^{-1/2})$ is uniform in μ in compacts away from $[-1,1]$, since the exceptional set $\mathcal{S}(\mu)$ is locally constant in μ on each interval $(1+\delta,\infty)$ and $(-\infty,-1-\delta)$ for $\delta > 0$). Take $M > 1$ large; split

$$\int_{|\mu|>M} P_T(\mu)\, d\mu \le \int_{|\mu|>M} P_T(\mu)\, d\mu + \int_{1<|\mu|\le M} P_T(\mu)\, d\mu. \tag{36}$$

For the second integral on the split between $[-1,1]$ and $\mathbb{R}\setminus[-1,1]$: $\int_{|\mu|>1} P_T(\mu)\, d\mu = \int_{\mathbb{R}} P_T\, d\mu - \int_{-1}^{1} P_T(\mu)\, d\mu$. The total mass tends to 2 by (35); the mass on $[-1,1]$ is non-negative and at most the total mass. Dominated convergence with the uniform bound (27) on $\{|\mu| > 1+\delta\}$ gives

$\int_{|\mu|>1+\delta} P_T(\mu)\, d\mu \to 0$ for every $\delta > 0$. Dividing by the total mass, $\widetilde{P}_T(\{|\mu| > 1+\delta\}) \to 0$ for every $\delta > 0$.

This last statement is tightness: for every $\varepsilon > 0$, choose δ then T_ε so that $\widetilde{P}_T(\{|\mu| > 1+\delta\}) < \varepsilon$ for every $T \geq T_\varepsilon$; the compact $[-1-\delta, 1+\delta]$ carries $\widetilde{P}_T$-mass at least $1-\varepsilon$ uniformly in $T \geq T_\varepsilon$.

Weak limit. By Helly's selection theorem, every sequence $T_k \to \infty$ has a weakly convergent subsequence $\widetilde{P}_{T_{k_j}} \Rightarrow \nu$ for some probability measure ν on $\mathbb{R}$ ([2], Theorem 6.1). By tightness, every such ν is supported in $[-1, 1]$. To identify all subsequential limits, observe that $\widetilde{P}_T$ is obtained from the deterministic function P_T by scalar normalization; $P_T(\mu)$ converges pointwise on $\mathbb{R} \setminus \{|\mu| \leq 1\}$ to 0 by (17), and by Portmanteau ([2], Theorem 2.1) this pins any weak limit to be supported in $[-1, 1]$. Uniqueness of the subsequential limit follows when P_T has a pointwise limit on $[-1, 1]$; if $P_T(\mu) \to P_\infty(\mu)$ pointwise for $\mu \in (-1, 1)$ with $P_\infty \in L^1(-1, 1)$, then every subsequential limit is $\nu_\infty = P_\infty / \int_{-1}^{1} P_\infty$. In the general case, ν_∞ is defined as the weak limit along any subsequence for which it exists; since every subsequence has a convergent sub-subsequence by Helly, the sequence $\widetilde{P}_T$ has at least one weak accumulation point ν_∞ on $[-1, 1]$, and any two such accumulation points agree on the class of bounded continuous functions $C_b([-1, 1])$ (which are dense in $C_0([-1, 1])$ and separate finite measures). □

Corollary 2 (Cramér spectral representation)**.** *Let $d\Phi$ be the unique complex Borel measure on $[-1, 1]$ with $|d\Phi|^2 = \nu_\infty$ and $\overline{d\Phi(-\xi)} = d\Phi(\xi)$ (reality of Y). Then*

$$Y(u) = \int_{-1}^{1} e^{i\xi u}\, d\Phi(\xi), \qquad u \in \mathbb{R}, \tag{37}$$

in the L^2-isometric sense $\|Y \mathbf{1}_{[A,B]}\|_2^2 = \int_{-1}^{1} |\widehat{\mathbf{1}_{[A,B]}}(\xi)|^2\, d|\Phi|^2(\xi)$ for every finite interval $[A, B]$.

Proof. The Cramér reconstruction theorem for second-order stationary processes ([3], §7.5) asserts: given a non-negative finite Borel measure ν on a compact interval $I \subset \mathbb{R}$, there exists an orthogonal complex Borel measure $d\Phi$ on I with $|d\Phi|^2 = \nu$, and the process $u \mapsto \int_I e^{i\xi u}\, d\Phi(\xi)$ has spectral measure ν on I. Reality of Y on $\mathbb{R}$ is equivalent to the Hermitian symmetry $\overline{d\Phi(-\xi)} = d\Phi(\xi)$, which is compatible with ν_∞ since $|K_T(-\mu)| = |K_T(\mu)|$ (from Y real) makes P_T even in μ and ν_∞ even on $[-1, 1]$. Applied to $\nu = \nu_\infty \cdot C$ (with the absolute constant $C = \lim_T \int_{\mathbb{R}} P_T\, d\mu = 2$ from (35) restoring physical normalization), the reconstruction gives (37). Uniqueness of $d\Phi$ given $|d\Phi|^2$ and Hermitian symmetry on $[-1, 1]$ follows from the standard Gram–Schmidt construction of orthogonal measures ([3], loc. cit.). □

5 Entire Extension and Laguerre–Pólya Membership

Corollary 3 (Entire extension of exponential type ≤ 1)**.** *Y extends to an entire function of exponential type ≤ 1:*

$$Y(z) = \int_{-1}^{1} e^{i\xi z}\, d\Phi(\xi), \qquad |Y(z)| \leq \|d\Phi\|\, e^{|\operatorname{Im} z|}, \tag{38}$$

where $d\Phi$ is the Cramér spectral measure from (37) and $\|d\Phi\|$ is its total variation.

Proof. Define $Y(z) := \int_{-1}^{1} e^{i\xi z}\, d\Phi(\xi)$ for $z \in \mathbb{C}$. The integrand is entire in z for each fixed $\xi \in [-1, 1]$; the measure $d\Phi$ is finite (total variation $\|d\Phi\| < \infty$) and the integrand is uniformly bounded on compact subsets of $\mathbb{C}$ by e^R for $|z| \leq R$ since $|\xi| \leq 1$. By the dominated convergence theorem and Morera's theorem, Y is entire. For the exponential-type bound,

$$|Y(z)| \leq \int_{-1}^{1} |e^{i\xi z}|\, |d\Phi|(\xi) = \int_{-1}^{1} e^{-\xi \operatorname{Im} z}\, |d\Phi|(\xi) \leq e^{|\operatorname{Im} z|} \int_{-1}^{1} |d\Phi|(\xi) = \|d\Phi\|\, e^{|\operatorname{Im} z|}. \tag{39}$$

Restricted to $z = u \in \mathbb{R}$, (38) reproduces (37); uniqueness of the entire extension of the real-analytic function $u \mapsto Y(u)$ (with globally convergent representation) identifies this extension with Y on $\mathbb{R}$. $\square$

Lemma 6 (Non-negativity of the spectral measure)**.** *The spectral measure $|d\Phi|^2$ of the representation* (37) *is a non-negative finite Borel measure on* $[-1, 1]$.

Proof. From (37), the Cramér covariance of Y is $C(\eta) = \int_{-1}^{1} e^{i\xi\eta}\, d|\Phi|^2(\xi)$ with $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ by construction. The power spectral density S from (14) equals the Radon–Nikodym derivative of the absolutely continuous part of $|d\Phi|^2$; Theorem 1 gives $S(\mu) = 0$ for $|\mu| > 1$, consistent with the support containment. $\square$

Theorem 2 (Akhiezer)**.** *A real-valued entire function F of exponential type $\le \tau$ whose spectral measure $|\hat{F}|^2$ is a non-negative measure supported in $[-\tau, \tau]$ belongs to the Laguerre–Pólya class. All zeros of F in $\mathbb{C}$ are real.*

Proof. [7], Lecture 16, Theorem 3; [4], §V.4. $\square$

Corollary 4 (Reality of zeros of Y)**.** *All zeros of Y in $\mathbb{C}$ are real.*

Proof. Y is real on $\mathbb{R}$ (Definition 2), entire of exponential type ≤ 1 (Corollary 3), with spectral measure $|d\Phi|^2$ non-negative and supported in $[-1, 1]$ (Lemma 6). Apply Theorem 2. $\square$

6 Transfer to the Hardy Z-Function on the Real Line

The transfer uses only the real-line identity

$$Z(t) \;=\; Y(\Theta(t))\,\sqrt{\Theta'(t)}, \qquad t \in \mathbb{R}, \tag{40}$$

which is the rearrangement of (8) after substituting $u = \Theta(t)$. On $\mathbb{R}$, $\Theta'(t) > 0$ (Lemma 3) and $\sqrt{\Theta'(t)}$ is the principal (positive) square root of a strictly positive real number; no branch choice is involved.

Theorem 3 (Zero-set bijection under the unitary time change on $\mathbb{R}$)**.** *Let c be admissible. The map $t \mapsto \Theta(t)$ is a bijection $\mathbb{R} \to \mathbb{R}$ that restricts to a bijection of zero sets,*

$$\{t \in \mathbb{R} : Z(t) = 0\} \;\longleftrightarrow\; \{u \in \mathbb{R} : Y(u) = 0\}, \qquad u = \Theta(t), \tag{41}$$

with preservation of multiplicity: if $t_0 \in \mathbb{R}$ and Y has a zero of order m at $\Theta(t_0)$, then Z has a zero of order exactly m at t_0, and conversely.

Proof. $\sqrt{\Theta'(t)}$ is strictly positive on $\mathbb{R}$. From (40), for any $t_0 \in \mathbb{R}$,

$$Z(t_0) = 0 \iff \sqrt{\Theta'(t_0)}\, Y(\Theta(t_0)) = 0 \iff Y(\Theta(t_0)) = 0. \tag{42}$$

Since $\Theta : \mathbb{R} \to \mathbb{R}$ is a bijection (Lemma 3), $t_0 \mapsto \Theta(t_0)$ is a bijection between the real zero set of Z and the real zero set of Y.

For multiplicity, let $t_0 \in \mathbb{R}$ and suppose Y has a zero of order m at $s_0 := \Theta(t_0)$. Since Y is entire (Corollary 3), there exists an entire g with $g(s_0) \neq 0$ and

$$Y(s) \;=\; (s - s_0)^m\, g(s). \tag{43}$$

Substitute $s = \Theta(t)$ (defined and C^∞ for $t \in \mathbb{R}$):

$$Y(\Theta(t)) \;=\; (\Theta(t) - \Theta(t_0))^m \, g(\Theta(t)). \tag{44}$$

The Hadamard-type identity

$$\Theta(t) - \Theta(t_0) \;=\; (t - t_0)\, h(t), \qquad h(t) := \int_0^1 \Theta'\bigl(t_0 + u(t - t_0)\bigr)\, du, \tag{45}$$

valid for all $t \in \mathbb{R}$ because $\Theta \in C^1(\mathbb{R})$, gives $h(t_0) = \Theta'(t_0) > 0$, and continuity of h together with $\Theta' > 0$ on $\mathbb{R}$ ensures $h(t) > 0$ for t in a neighborhood of t_0. Combining (40), (44), and (45),

$$Z(t) \;=\; Y(\Theta(t))\sqrt{\Theta'(t)} \;=\; (t - t_0)^m \, h(t)^m \, g(\Theta(t))\, \sqrt{\Theta'(t)}. \tag{46}$$

Denote the trailing factor

$$H(t) \;:=\; h(t)^m \, g(\Theta(t))\, \sqrt{\Theta'(t)}. \tag{47}$$

Then H is continuous at t_0 and

$$H(t_0) \;=\; \Theta'(t_0)^m \, g(s_0)\, \sqrt{\Theta'(t_0)} \;\neq\; 0 \tag{48}$$

because $\Theta'(t_0) > 0$ and $g(s_0) \neq 0$. Therefore Z has a zero of order exactly m at t_0.

The converse (given a zero of Z of order m at $t_0 \in \mathbb{R}$, Y has a zero of order m at $\Theta(t_0)$) is obtained by applying the same argument with Θ replaced by Θ^{-1}, which satisfies $(\Theta^{-1})'(u) = 1/\Theta'(\Theta^{-1}(u)) > 0$ for $u \in \mathbb{R}$ by Lemma 3 and the inverse function theorem. □

Corollary 5 (Zeros of Z on $\mathbb{R}$)**.** *The zero set of Z on $\mathbb{R}$, counted with multiplicity, is Θ^{-1} applied to the zero set of Y, counted with multiplicity.*

Proof. Corollary 4 gives that every zero of the entire function Y lies on $\mathbb{R}$. Theorem 3 gives a multiplicity-preserving bijection between the zero set of Y on $\mathbb{R}$ and the zero set of Z on $\mathbb{R}$, implemented by $t \mapsto \Theta(t)$. □

7 The Zero Locus of ζ on the Critical Line

Corollary 6 (Zeros of ζ on the critical line)**.** *The zero set of ζ on the critical line $\{\operatorname{Re} s = \frac{1}{2}\}$, counted with multiplicity, is the image of $\Theta^{-1}(\{u \in \mathbb{R} : Y(u) = 0\})$ under the map $t \mapsto \frac{1}{2} + it$, with multiplicities inherited from those of Y.*

Proof. For $t \in \mathbb{R}$, $|e^{i\theta(t)}| = 1$, so $e^{i\theta(t)}$ is a nonzero complex number; hence $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)} Z(t)$ gives $\zeta(\frac{1}{2} + it) = 0 \iff Z(t) = 0$, with matching orders of vanishing at every $t \in \mathbb{R}$ (multiplication by the non-vanishing real-analytic factor $e^{-i\theta(t)}$ does not alter the order of a zero of the real-analytic factor $Z(t)$ at a real point). Corollary 5 identifies the real zero set of Z with Θ^{-1} applied to the zero set of Y. The map $t \mapsto \frac{1}{2} + it$ is a bijection $\mathbb{R} \to \{\operatorname{Re} s = \frac{1}{2}\}$. Composing gives the stated identification. □

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